

SIMATS Engineering

Saveetha Institute of Medical and Technical Sciences

CO1_AT2 — Scenario-Based Research Assessment

Domain: Healthcare — AI in Robotic Surgery

1. Title of AI Application

da Vinci 5 — AI-Integrated Robotic-Assisted Surgical System (Intuitive Surgical). This report looks at the latest generation of AI-enabled robotic surgery platforms, with brief comparison to Medtronic's Hugo RAS and the Johns Hopkins research systems STAR / SRT-H. Focus area: Healthcare — Surgical Robotics.

2. Domain Selection

I chose Healthcare, and more specifically robotic-assisted and early autonomous surgery, because it is one of the hardest places to apply AI well. The system has to combine computer vision, force sensing, real-time control and machine learning, all under strict safety rules and regulatory oversight. Mistakes are not theoretical — they affect patients directly — so the engineering bar is high.

3. Introduction and Background

Robotic surgery started as tele-operated machines. The surgeon sat at a console and the robot simply copied hand movements into smaller instrument motions inside the patient. Over the last ten years or so, AI has been added on top of that mechanical base. The platforms are no longer only passive tools; they can give feedback, analyse technique and, in research settings, handle some repetitive steps on their own.

Intuitive Surgical's da Vinci line has been used in millions of procedures. The fifth generation, da Vinci 5, received FDA clearance in 2024 and brings AI-based force feedback, better 3D vision and software that studies how surgeons operate across cases. Other systems are catching up. Medtronic's Hugo RAS was cleared for certain urologic procedures in late 2025, and CMR Surgical's Versius Plus is also active in the market. On the research side, Johns Hopkins' STAR and the later SRT-H system have shown that limited autonomous tasks are possible in controlled conditions. Taken together, these examples show how AI is changing efficiency, precision and training in the operating room.

Fig. 1 — Typical layout of an AI-assisted robotic surgery suite

(Schematic drawn for this report; not a product photo)

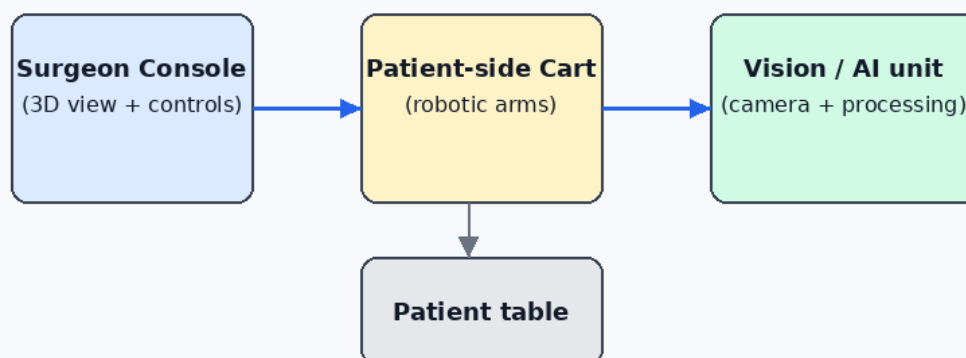


Fig. 1: Simple schematic of an AI-integrated robotic surgery setup (drawn for this report).

Fig. 1a — Core elements of AI-assisted robotic surgery



Fig. 1a: Main pieces of the system — robotic arms, console and patient area (illustration).

4. Problem Identification

Open surgery and even standard keyhole surgery still run into basic human limits: hand tremor, tiredness during long cases, different technique from one surgeon to another, limited depth perception, and a shortage of trained specialists. That shortage is expected to get worse in areas such as ophthalmology and general surgery. Complication rates, longer operating times and uneven results between hospitals are still common. Training new surgeons to a steady standard takes a long time and a lot of resources, and advanced expertise is not evenly available around the world.

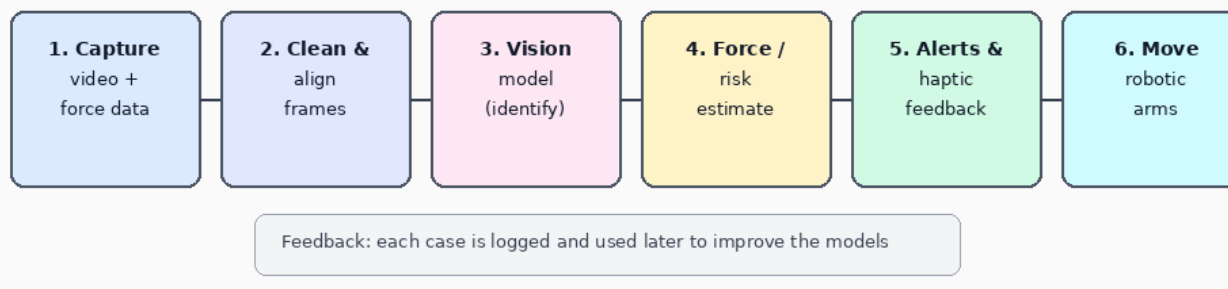
5. Need for AI Solution

Earlier robotic systems already cut tremor and improved dexterity, but they still needed constant human control and gave almost no intelligent feedback about tissue force, anatomy or how well the technique was going. AI is useful here for a few clear reasons. Computer vision can pick out anatomical structures in real time and help avoid injury. Models that estimate force can warn the surgeon when pressure is getting high enough to damage tissue. Analytics tools can compare a surgeon's movements with data from thousands of earlier cases and point out where consistency can improve. AI also makes it realistic to automate well-defined, repetitive steps such as suturing, which can reduce fatigue on long procedures — something pure rule-based automation struggled with because human anatomy varies so much.

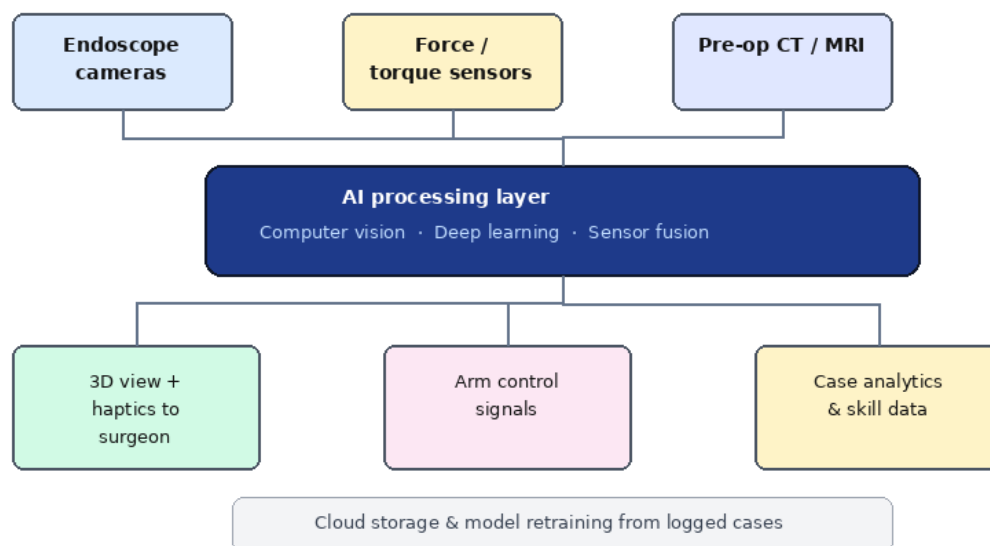
6. AI Technologies Used

Current AI-enabled surgical robots usually combine several pieces of technology:

- **Computer vision** — real-time recognition of tissue types, organs and instruments from the endoscopic camera, used for scene understanding and safety alerts.
- **Machine learning / deep learning** — models trained on large libraries of recorded procedures. These guide implant placement (for example MAKO's 3D joint modelling for knee and hip replacement) and power case-analytics and skill-assessment tools.
- **Reinforcement learning** — used in research systems such as SRT-H to learn sub-tasks (suturing, tissue handling) from demonstration and correction.
- **Sensor fusion and force-feedback AI** — haptic sensors plus AI processing turn tissue resistance into feedback the surgeon can feel, which helps limit excess force.
- **Robotics and control systems** — multi-arm platforms (da Vinci 5, Hugo RAS, Versius) that execute very precise movement based on the fused sensor and vision data.

Fig. 3 — How sensor data becomes a safe instrument movement*Fig. 3: Stages that turn raw sensor input into a safe surgical action.*

7. Working Mechanism

Fig. 2 — Overall system architecture (from sensors to action)*Fig. 2: System architecture — from sensors through AI processing to actuation and analytics.*

During an operation the system collects two main streams: video from the stereoscopic endoscopic cameras and force/torque readings from the instrument arms. On-board vision and machine-learning models process that stream to identify anatomy, track instrument position and estimate tissue resistance. The results feed two kinds of output. First, an enhanced 3D view and haptic force signal go back to the surgeon at the console, supporting tremor-free, sub-millimetre control. Second, case data is logged for later review; that data is used to refine the models and to give the surgical team performance feedback. In research systems such as STAR/SRT-H the same pipeline is extended so the model itself can plan and carry out a defined sub-task under close human supervision, rather than only assisting a human-driven instrument.

8. Data Sources and Processing

The models are trained mainly on recorded surgical video and kinematic data. Intuitive Surgical alone has reported logging well over two million da Vinci procedures in a recent year. Pre-operative CT and MRI scans are also used to build 3D anatomical models before surgery. Processing usually starts with clinical experts annotating surgical phases and landmarks. Supervised training of vision and prediction models follows, and reinforcement learning is applied in laboratory settings for autonomous sub-tasks. New procedures keep adding data, so models can be retrained over time. That continuous collection also creates strict requirements around privacy and governance, because surgical video is highly sensitive and often identifiable.

9. Real-World Implementation

Fig. 4 — Where robotic surgery sits in a hospital's AI tools



Fig. 4: Rough place of AI-enabled robotic surgery inside a hospital's wider AI toolkit.

Fig. 5 — Conceptual illustration of a clinical support robot

(Drawn for this report; not a commercial product photo)

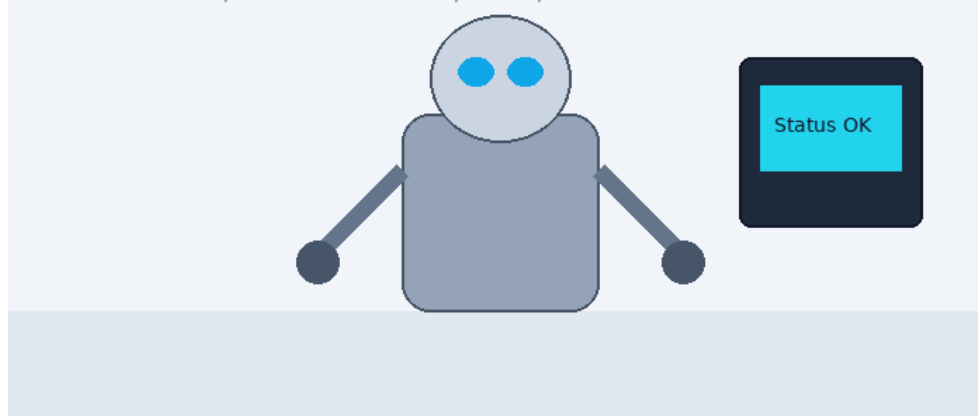


Fig. 5: Conceptual sketch of a clinical support robot (illustration only).

da Vinci systems are already in thousands of hospitals and are used in urology, gynaecology, general, cardiac and colorectal surgery. In 2026 UC San Diego Health reported the first West Coast spine cases with a new AI-powered robotic system that combines advanced imaging and guidance, showing the technology moving into neurosurgery as well. Medtronic's Hugo RAS, cleared in December 2025 for urologic procedures such as prostatectomy and nephrectomy, is active in more than 30 countries and includes an AI-based surgical-video analytics platform. Stryker's MAKO system is widely used for AI-guided joint replacement. Platforms with telesurgery capability, for example Japan's hinotori, are expanding access in Asia for gynaecology and general surgery.

10. Impact and Benefits

Recent reviews of AI-driven robotic surgery report meaningful clinical gains, though the exact numbers vary by study and procedure. Roughly speaking, published work has described about a 25% reduction in operative time, around a 30% drop in intraoperative complications, improvements in precision on the order of 40%, and recovery times that can be about 15% shorter than with purely manual techniques. Surgeon workflow tends to improve and overall costs show modest reductions in some settings. Joint-replacement studies with AI-guided robots have shown better implant positioning accuracy. Force-feedback technology has been shown to reduce the force applied to tissue — a safety-relevant change even while its effect on hard clinical endpoints is still being studied.

11. Challenges and Limitations

Progress is real, but important limits remain. A 2026 evidence review notes that a clear clinical benefit of AI force feedback has not yet been statistically proven in pilot trials. Fully autonomous systems have so far been tested mainly in laboratory and animal settings, not in routine human care. Real-time clinical use of

computer-vision tools is still limited, and external validation across different hospitals is weak. Accountability is an open ethical question when an AI-assisted decision contributes to a bad outcome. The platforms themselves are expensive (often multi-million-dollar systems), which raises concerns about access in lower-resource settings. Privacy and security of recorded surgical video are additional regulatory and technical challenges.

12. Future Scope and Emerging Trends

The direction of the field looks like a gradual, carefully supervised move from pure assistance toward what some authors call "augmented dexterity": well-defined, repetitive sub-tasks such as suturing are automated under the surgeon's close watch, rather than replacing the surgeon. Telesurgery is another active frontier — a specialist operating remotely on a patient in another location — which could help with the uneven global distribution of surgical skill. Narrower robots are also appearing; ForSight Robotics, for example, is building a platform aimed at cataract surgery in response to projected shortages of ophthalmic surgeons. Work on microrobots, better validation of force feedback, and larger multi-centre trials will decide how far autonomy can safely go over the next several years.

13. Conclusion

AI-integrated robotic surgery, illustrated by platforms such as da Vinci 5 and its competitors, shows that artificial intelligence can usefully extend human surgical skill — better precision, fewer complications, shorter recovery — without taking over clinical judgement. The important pattern is incremental and safety-conscious: first tremor reduction and 3D visualisation, then AI-guided implant planning and force feedback, then early, tightly supervised autonomous sub-tasks. As more clinical evidence appears and regulators catch up, this area is in a strong position to widen access to high-quality surgery and to help address the shortage of specialist surgeons.

14. References

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